Daily Rant Material Strength and Simulation

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- **Introduction to Matrices:** How to compute the inverse of a matrix.

$$\mathbf{l_1} : y = a_1x + b_1 \rightarrow y - a_1x = b_1$$
$$\mathbf{l_2} : y = a_2x + b_2 \rightarrow y - a_2x = b_2$$
$$\begin{bmatrix} -a_1 & 1 \\ -a_2 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{bmatrix} b1 \\ b2 \end{bmatrix}$$

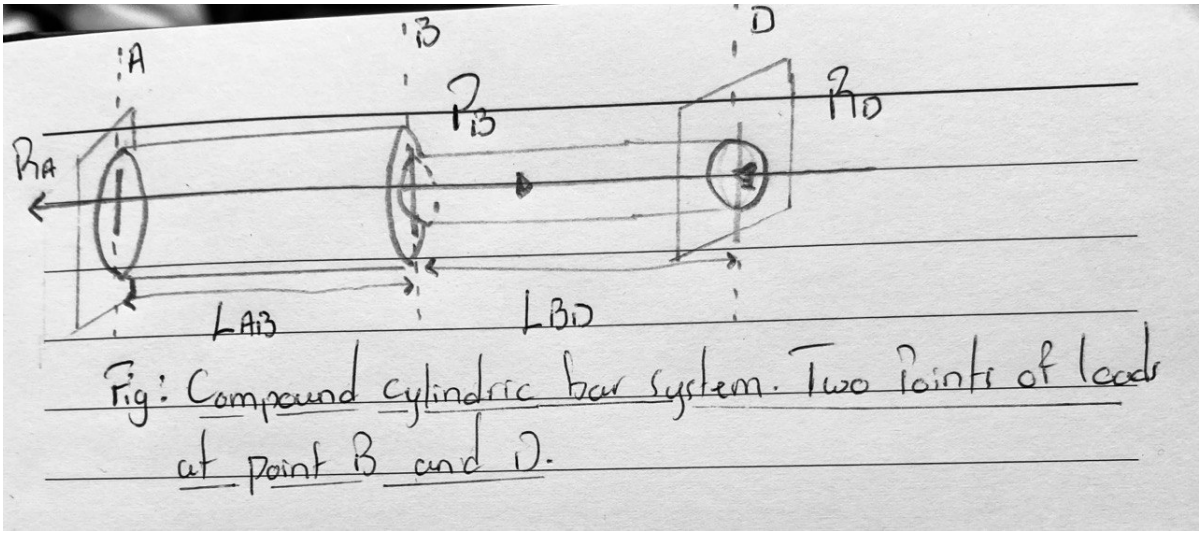
Inverse

$$\Big[A\Big]\Big[A\Big]^{-1} = \Big[A\Big]^{-1}\Big[A\Big] = \Big[I\Big]$$

We worked on forces namely internal forces on a bar and static equilibrium. Below is a diagram showing the different loads/forces acting on a bar at fixed on both ends. We used 3 different approaches in solving for the **stress** and **deformation**. Stress and deformation are given resolved using the formulas below:

$$\sigma = \frac{N}{A} \qquad \text{Stress}$$
$$A = \frac{\pi d^2}{4} \qquad \text{Cross sectional Area}$$
$$\delta = \sum \frac{F \cdot L}{E \cdot A} \qquad \text{Deformation}$$

Below is a diagram showing all the forces acting on compound bar.



Analytical: The problem was approached analytically using the principles of classical mechanics, specifically static and solid mechanics.
Numerical: The numerical method was developed to simulate the physical system and solve the unknown stresses and deformations using matrices.
Kraken Program: Finally a specialised model built using Kraken(Program from the professor himself). The program is capable of computing the resulting forces on the system and even drawing the physical system itself.